

Abstract No.

Comprehensive analysis of TERT promoter methylation in glioblastoma: associations with molecular features, expression, and survival

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While TERT promoter mutations are a well-known molecular feature of glioblastoma, the relevance of TERT promoter methylation remains unclear. Recent studies suggest that TERT promoter methylation is associated with survival in other malignancies. Yet, a comprehensive analysis of TERT promoter methylation and its potential molecular and clinical relevance in glioblastoma is missing. We analyzed 301 glioblastomas from three public datasets (TCGA, GSE195684, GSE60274) and 121 cases from our diagnostic routine. Mutational status, including TERT promoter mutations, MGMT promoter status and copy number variation (CNV) profiles were assessed. Beta values of TERT promoter-associated CpG sites were extracted from Illumina 450k and EPICv1 arrays. TERT promoter methylation was compared with normal brain tissue using GSE90496. TERT RNA expression (TCGA) was correlated with promoter methylation using Spearman's rank correlation. Differential methylation was analyzed with limma, and survival with survminer.

In-house data showed no significant correlation between TERT promoter mutations and methylation, but a trend toward cg11625005 hypermethylation in TERT wild-type tumors. Using Spearman's rank correlation TERT locus CNV moderately negatively correlated with methylation at cg7380026 ($\rho = -0.2$) and cg23250191 ($\rho = -0.2$). TERT promoter CpG sites were significantly differentially methylated compared to normal brain tissue. In TCGA data, methylation of cg10896616 ($\rho = 0.4$, adj. $p < 0.05$), cg17166338 ($\rho = 0.3$, adj. $p < 0.05$), and cg26006951 ($\rho = 0.3$, adj. $p < 0.05$) positively correlated with TERT expression. Methylation of cg10896616 was associated with overall survival in a multivariate Cox regression model adjusted for MGMT status, age, and sex (HR = 1.1, $p < 0.05$). Differential methylation using cg10896616 as a continuous variable shows gene enrichments in cancer associated pathways, e.g. MAPK, mTOR and Wnt pathway.

TERT promoter methylation has potentially relevant biological and clinical implications in glioblastoma and may improve diagnostic precision and understanding of tumor biology.



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